NuPa category reorganization

Discussion draft | 27 October 2026
Scope: the former Collider Physics section

Purpose. Group concrete studies by the particles or reactions under investigation, and reserve a clearly defined category for methods applicable across reactions. The revised parent name also accommodates low-energy photon scattering and annihilation.

Comparison. Pre-trial README at commit **6d4eaf9** versus the current working draft. The same **125 unique papers** are retained; **17 change primary category**. Papers affected only by a category-name change are excluded from the lists on pages 2-3.

Category names and paper counts

Parent: Collider Physics → Particle Scattering and Decays. Top Quark Sector and Electroweak Bosons and Higgs keep their subcategory names.

Previous subcategory	Revised subcategory	Before	After
Top Quark Sector	Top Quark Sector	29	29
Electroweak Bosons and Higgs	Electroweak Bosons and Higgs	33	30
Flavor and Hadron Spectroscopy	Quarks and Hadrons	36	34
New category	Photon and Lepton Processes	-	15
Observables and Methodology	General Methods	27	17

Why these changes?

Broader parent scope. Compton scattering and positronium decay belong naturally with particle scattering and decays, including reactions outside high-energy colliders.

Quarks and Hadrons. The contents include quark flavor structure, neutral-meson mixing, and baryon spin and decay studies. “Spectroscopy” suggests a stronger emphasis on mass spectra than the list has. “Flavor” names a property and a research theme; the revised label names the physical systems.

Photon and Lepton Processes. A dedicated home for concrete photon-polarization, Compton, positronium, and tau-pair studies prevents these systems from being treated as miscellaneous methodology.

General Methods. Keep process-independent reconstruction, observable design, inference frameworks, and general reviews together. A specific process used as an example does not by itself make a paper system-specific.

Placement rule

Choose the primary category from the central physical system or the generality of the contribution. Higgs-decay and electroweak-coupling studies stay in EW/Higgs; studies of the tau-pair system itself go to Photon and Lepton Processes. Cross-listing retains relevant applications.

Status: a trial organization in README and the category descriptions. Synchronization with the canonical paper data and regeneration of the living-review PDF remain pending.

Changes of primary category

Three studies of tau pairs and two papers on general methods.
Subcategory names below omit the common parent prefix.

A. Tau-pair studies → Photon and Lepton Processes (3)

Shared reason: the central object is the tau-pair spin and decay system, including its kinematic reconstruction. These contributions belong together as studies of the lepton system.

1. Prospects of measuring quantum entanglement in ττ final states at a future e+e- Higgs factory
arXiv:2409.20239 | INSPIRE 2835996

From: Electroweak Bosons and Higgs

2. Probing entanglement and testing Bell inequality violation with e+e- → τ+τ- at Belle II
arXiv:2311.17555 | INSPIRE 2727866

From: Flavor and Hadron Spectroscopy

3. Experimental aspects of the Quantum Tomography of tau lepton pairs at a Higgs factory collider
arXiv:2603.12609 | INSPIRE 3129062

From: Flavor and Hadron Spectroscopy

B. EW/Higgs → General Methods (2)

Shared reason: these papers develop inference or spin-observable frameworks with applicability beyond a particular weak-boson or Higgs channel. EW/Higgs applications remain visible through cross-listing.

4. Quantum state tomography, entanglement detection and Bell violation prospects in weak decays of massive particles
arXiv:2209.13990 | INSPIRE 2157995

5. SMEFT Constraints via Quantum Entanglement in Portal-Mediated Scattering
arXiv:2610.09751 | INSPIRE 3212551

Additional cross-listing changes

Six secondary-category links were added for five papers. These are not additional primary relocations. Existing cross-lists were retained.

Already listed above: paper 4 gains cross-lists in Top Quark Sector and EW/Higgs; paper 5 gains an EW/Higgs cross-list.

Primary unchanged; added to Photon and Lepton Processes:

C1. Quantum tomography with τ leptons at the FCC-ee
arXiv:2405.09201 | INSPIRE 2787034

Primary: EW/Higgs. The Z-pole electroweak-coupling focus is retained; the tau-system application motivates the added cross-list.

C2. Quantum tomography for collider physics: Illustrations with lepton pair production
arXiv:1707.01638 | INSPIRE 1608880

C3. Entanglement Detection with Rotationally Covariant Measurements -- From Compton Scattering to Lemonade
arXiv:2512.03984 | INSPIRE 3089176

Shared reason for C2-C3: retain General Methods as primary because the contribution is a general inference or measurement framework; add visibility under its lepton/photon applications.

Studies of photon systems

Observables and Methodology → Photon and Lepton Processes
12 changes of primary category

Shared reason: these papers study photon production, annihilation, scattering, or polarimetry in specific physical systems. They therefore move to Photon and Lepton Processes; General Methods is reserved for results intended to apply across reactions.

C. High-energy photon pairs and polarimetry (2)

6. Quantum entanglement of photon pairs at proton-proton colliders
arXiv:2607.00971 | INSPIRE 3176266

7. High Energy Photon Polarimetry at Lepton Colliders: Quantum Information from Converted Photons
arXiv:2607.08816 | INSPIRE 3179608

D. Positronium and annihilation photons (3)

8. Potential of the J-PET detector for studies of discrete symmetries in decays of positronium atom - a purely leptonic system
arXiv:1602.05226 | INSPIRE 1422075

9. Genuine Multipartite Entanglement in the 3-Photon Decay of Positronium
arXiv:1706.06505 | INSPIRE 1606137

10. Testing CPT symmetry in ortho-positronium decays with positronium annihilation tomography
arXiv:2112.04235 | INSPIRE 1955404

E. Compton-scattering systems (7)

11. Witnessing Entanglement In Compton Scattering Processes Via Mutually Unbiased Bases
arXiv:1807.04934 | INSPIRE 1682310

12. Probing entanglement in Compton interactions
arXiv:1910.05537 | INSPIRE 1758833

13. Entanglement Entropy of Compton Scattering with a Witness
arXiv:2305.10027 | INSPIRE 2660572

14. Measuring the Evolution of Entanglement in Compton Scattering
arXiv:2406.14352 | INSPIRE 2800856

15. Compton scattering mediated by quantum entanglement
DOI: 10.1103/PhysRevA.111.053708 | INSPIRE 2919423

16. Entanglement, steering, and separability in Compton-scattered annihilation photons
DOI: 10.1103/j7p7-ckp9 | INSPIRE 2968534

17. Photon-nucleon entanglement in Compton scattering at low and high energies
arXiv:2608.05330 | INSPIRE 3188079